

PROJECT: Multi-Region Disaster Recovery Plan

Services: EC2, RDS, Route 53, CloudFormation

Description: Set up a disaster recovery architecture where data and applications are replicated across two regions. Automate failover using Route 53 health checks.

Skills: High availability, failover planning, cross-region replication.

Definition: This project implements a Multi-Region Disaster Recovery architecture on AWS to ensure application availability during failures. The application is deployed in a primary region (Mumbai) and replicated to a secondary region (Virginia). AWS Route 53 health checks and failover routing automatically redirect traffic to the backup server when the primary server becomes unavailable. This architecture improves high availability, reliability, and business continuity.

Scope of this project: The scope of this project is to design and implement a highly available and fault-tolerant cloud architecture using AWS services, where application and database resources are deployed across multiple AWS regions to ensure business continuity during failures. It includes setting up primary and secondary environments in different regions, enabling automated failover using health checks and DNS routing, and implementing cross-region data replication for data durability. The project covers deployment of compute resources using EC2, database management using RDS with replication, and traffic routing using Route 53 for failover handling. It also involves monitoring system health, simulating disaster scenarios, and validating automatic recovery to ensure minimal downtime and high availability of the application.

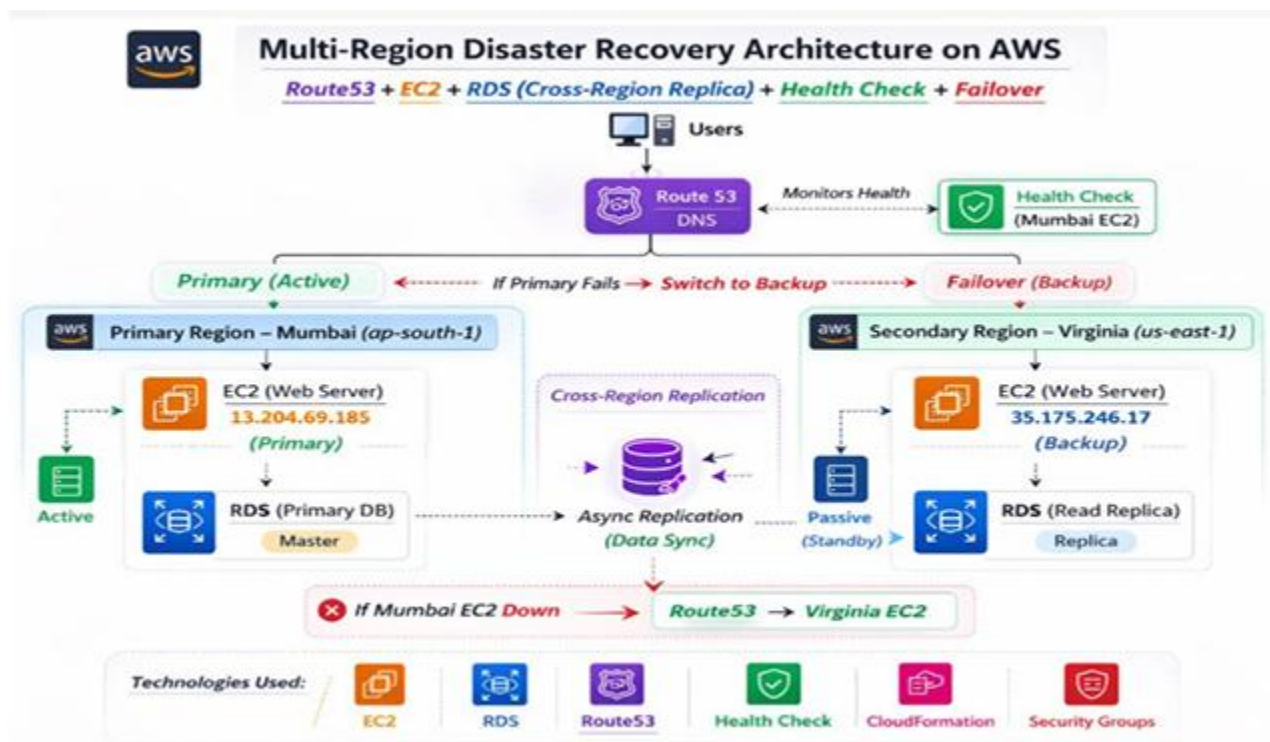
Methodologies:

- Deploy application in two regions (primary Mumbai and secondary Virginia) using EC2 for high availability.
- Use Amazon RDS cross-region read replica to keep database synchronized between regions.
- Implement Route 53 failover routing policy to automatically redirect traffic during failures.
- Configure health checks to continuously monitor the primary region's availability.
- Ensure secure communication using VPC, subnets, and security groups across regions.
- Perform failover testing and monitoring to validate disaster recovery and system resilience.

Services Used in this Project:

- Amazon EC2 – Used to host application servers in primary (Mumbai) and secondary (Virginia) regions.
- Amazon RDS – Used to store application data with cross-region read replica for disaster recovery.
- Amazon Route 53 – Used for DNS management and automatic failover routing between regions.
- Amazon VPC – Used to create isolated network environments for secure communication.
- AWS Identity and Access Management (IAM) – Used to manage access permissions and secure AWS resources.
- Amazon CloudWatch – Used to monitor logs, health checks, and system performance.

Architecture Diagram:



STEP 1: Launch EC2 Instance in Primary Region (Mumbai)

- ## STEP 2: Install Web Application on Mumbai EC2

```
AWS [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) Shreenivasa Shreenivasa
```

```
Complete!  
[root@ip-172-31-1-70 ec2-user]# yum install httpd -y  
Last metadata expiration check: 0:00:08 ago on Mon Apr 20 16:41:34 2026.  
Dependencies resolved.
```

Package	Architecture	Version	Repository	Size
Installing:				
httpd	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	47 k
Installing dependencies:				
apr	x86_64	1.7.5-1.amzn2023.0.4	amazonlinux	129 k
apr-util	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	97 k
apr-util-ldap	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	13 k
generic-logos-httpd	noarch	18.0.0-12.amzn2023.0.3	amazonlinux	19 k
httpd-core	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	1.4 M
httpd-filesystem	noarch	2.4.66-1.amzn2023.0.1	amazonlinux	13 k
httpd-tools	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	81 k
libbrotli	x86_64	1.0.9-4.amzn2023.0.2	amazonlinux	315 k
mailcap	noarch	2.1.49-3.amzn2023.0.3	amazonlinux	33 k
Installing weak dependencies:				
apr-util-openssl	x86_64	1.6.3-1.amzn2023.0.2	amazonlinux	15 k
mod_http2	x86_64	2.0.27-1.amzn2023.0.3	amazonlinux	166 k
mod_lua	x86_64	2.4.66-1.amzn2023.0.1	amazonlinux	60 k

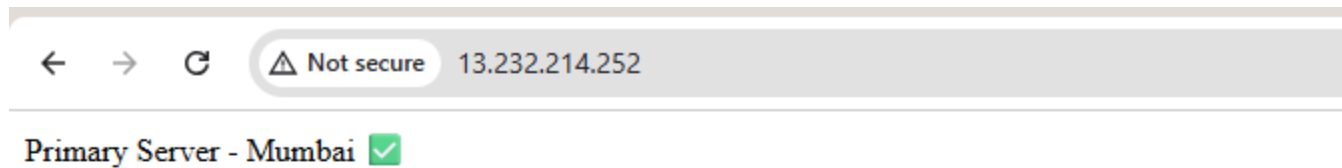
STEP 3: Installing and Configuring Apache Web Server on EC2 Instance

```
aws Search [Alt+S] Ask Amazon Q Asia Pacific (Mumbai) Shreenivasa (0741-3904-  
Shr  
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch httpd-2.4.66-1.amzn2023.0.1.x86_64  
httpd-core-2.4.66-1.amzn2023.0.1.x86_64 httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch httpd-tools-2.4.66-1.amzn2023.0.1.x86_64  
libbrotli-1.0.9-4.amzn2023.0.2.x86_64 mailcap-2.1.49-3.amzn2023.0.3.noarch mod_http2-2.0.27-1.amzn2023.0.3.x86_64  
mod_lua-2.4.66-1.amzn2023.0.1.x86_64  
Complete!  
[root@ip-172-31-1-70 ec2-user]# systemctl start httpd  
[root@ip-172-31-1-70 ec2-user]# systemctl enable httpd  
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
```

STEP 4: Create a web page

```
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.  
[root@ip-172-31-1-70 ec2-user]# echo "Primary Server - Mumbai" | sudo tee /var/www/html/index.html  
Primary Server - Mumbai
```

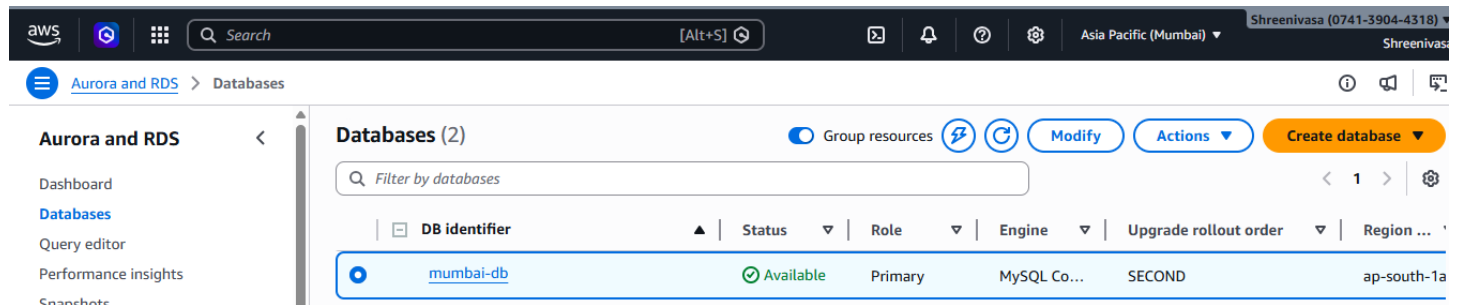
STEP 5: Now open the browser:



STEP 6: Create RDS Database in Mumbai Region

Go to AWS Console → RDS.

- Click Create Database



STEP 7: Create DB Subnet Group in Virginia

- Switch region to US East (N. Virginia).
- Go to RDS → Subnet Groups.
- Click Create DB Subnet Group.
- Select: > VPC > Two private subnets
- Create subnet group.

Aurora and RDS > Subnet groups

Subnet groups (1)

Filter by subnet group

☐	Name	Description	Status	VPC
☐	virginia-subnet-group	for multi region-project	Complete	vpc-06753b6960ba059f7

STEP 8: Create Cross-Region RDS Read Replica

- Go to RDS → Databases.
- Select the mumbai db.
- Click: ➤ Actions → Create cross-region replica
- Change region to: ➤ US East (N. Virginia)
- Select the Virginia subnet group.
- Create replica.

Aurora and RDS > Databases

Databases (1)

Group resources

Filter by databases

☐	DB identifier	Status	Role	Engine	Upgrade rollout order	Region ...
☒	virginia-replica	Available	Replica	MySQL Co...	SECOND	us-east-1b

STEP 9: Launch Backup EC2 in Virginia Region

EC2 > Instances

Instances (1/1) Info

Connect Instance state Actions Launch instances

Find Instance by attribute or tag (case-sensitive)

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
☒	backup-virgini...	i-0ff5a6cd9a73dba8f	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a

STEP 10: Install web server

```
aws  [Alt+S]  United States (N. Virginia)  Shreenivasa (0741-3904-4318)  Shreenivasa

#
##### Amazon Linux 2023
#####
#####
#/ https://aws.amazon.com/linux/amazon-linux-2023
V~' ->
~m/

[ec2-user@ip-172-31-3-135 ~]$ sudo su
[root@ip-172-31-3-135 ec2-user]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository                267 kB/s | 31 kB    00:00
Dependencies resolved.
Nothing to do.
Complete!
```

```

aws  [AWS] Search [Alt+S] [Icons] United States (N. Virginia) Shreenivasa (0741-3904-4318) Shreenivasa

[root@ip-172-31-3-135 ec2-user]# yum install httpd -y
Last metadata expiration check: 0:00:04 ago on Mon Apr 20 17:36:24 2026.
Dependencies resolved.

=====
Package                                Architecture      Version           Repository        Size
=====
Installing:
httpd                                  x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      47 k
Installing dependencies:
apr                                    x86_64            1.7.5-1.amzn2023.0.4   amazonlinux      129 k
apr-util                              x86_64            1.6.3-1.amzn2023.0.2   amazonlinux      97 k
apr-util-ldap                         x86_64            1.6.3-1.amzn2023.0.2   amazonlinux      13 k
generic-logos-httpd                  noarch            18.0.0-12.amzn2023.0.3  amazonlinux      19 k
httpd-core                            x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      1.4 M
httpd-filesystem                     noarch            2.4.66-1.amzn2023.0.1  amazonlinux      13 k
httpd-tools                           x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      81 k
libbrotli                             x86_64            1.0.9-4.amzn2023.0.2   amazonlinux      315 k
mailcap                               noarch            2.1.49-3.amzn2023.0.3  amazonlinux      33 k
Installing weak dependencies:
apr-util-openssl                      x86_64            1.6.3-1.amzn2023.0.2   amazonlinux      15 k
mod_http2                             x86_64            2.0.27-1.amzn2023.0.3  amazonlinux      166 k
mod_lua                               x86_64            2.4.66-1.amzn2023.0.1  amazonlinux      60 k
=====
Transaction Summary

```

STEP 11: Create backup page

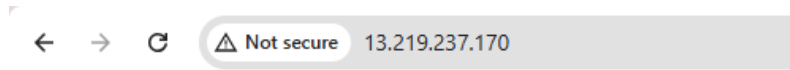
```

1.6.3-1.amzn2023.0.2.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64      generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      httpd-2.4.66-1
.amzn2023.0.1.x86_64
httpd-core-2.4.66-1.amzn2023.0.1.x86_64           httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch           httpd-tools-2.
4.66-1.amzn2023.0.1.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64              mailcap-2.1.49-3.amzn2023.0.3.noarch                    mod_http2-2.0.
27-1.amzn2023.0.3.x86_64
mod_lua-2.4.66-1.amzn2023.0.1.x86_64

Complete!
[root@ip-172-31-3-135 ec2-user]# systemctl start httpd
[root@ip-172-31-3-135 ec2-user]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-172-31-3-135 ec2-user]# sudo vi /var/www/html/index.html
[root@ip-172-31-3-135 ec2-user]# sudo systemctl restart httpd

```

STEP 12: Test in Browser



Backup Server - Virginia

STEP 13: Create Route 53 Hosted Zone

- Go to Route 53 → Hosted Zones.
- Click Create Hosted Zone.

The screenshot shows the AWS Route 53 console for the hosted zone 'dr-demo.com'. The left sidebar contains navigation links for Route 53, Global Resolver, and VPC Resolver. The main content area displays the 'Hosted zone details' for 'dr-demo.com', which is in 'Public' status. Below this, there are tabs for 'Records (4)', 'Accelerated recovery', 'DNSSEC signing', and 'Hosted zone tags (0)'. The 'Records (4)' tab is active, showing a table of DNS records. The table has columns for Record ID, Type, Routing type, Difference, Alias, Value/Route traffic to, TTL, and Health status. Two records are visible: an NS record for 'dr-demo.c...' and a SOA record for 'dr-demo.c...'. The NS record points to 'ns-1967.awsdns-53.co.uk' with a TTL of 172800. The SOA record points to 'ns-1967.awsdns-53.co.uk' with a TTL of 900.

Record ...	Type	Routin...	Differ...	Alias	Value/Route traffic to	TTL (s...	Health
☐ dr-demo.c...	NS	Simple	-	No	ns-1967.awsdns-53.co.uk. ns-1195.awsdns-21.org. ns-986.awsdns-59.net. ns-339.awsdns-42.com.	172800	-
☐ dr-demo.c...	SOA	Simple	-	No	ns-1967.awsdns-53.co.uk. a...	900	-

STEP 14: Create Route 53 Health Check

- Go to Route 53 → Health Checks.
- Click Create Health Check.

The screenshot shows the AWS Route 53 console for 'Health checks (2)'. The left sidebar contains navigation links for Route 53, Global Resolver, and VPC Resolver. The main content area displays the 'Health checks (2)' page, which includes a search bar and a table of health checks. The table has columns for ID, Name, Details, Status in last 24 hours, Current status, and Actions. One health check is visible with ID '1d3e546f...', Name 'mumbai-h...', and Details 'http://13...'. The status is 'Healthy'.

ID	Name	Details	Status in last 24 hours	Current s...	Actions
☐ 1d3e546f...	mumbai-h...	http://13...		Healthy	

STEP 15: Create Route 53 Failover Records

- Go to Hosted Zone → dr-demo.local.

The screenshot shows the AWS Route 53 console for the hosted zone 'dr-demo.local'. The left sidebar contains navigation links for Route 53, Global Resolver, and VPC Resolver. The main content area displays the 'Records (1/4)' page, which includes a search bar and a table of DNS records. The table has columns for Record ID, Type, Routing type, Difference, Alias, and Value. One record is visible with ID 'app.dr-de...', Type 'A', Routing type 'Failover', Difference 'Primary', and Value '13.232.214.252'.

Record ...	Type	Routin...	Differ...	Alias	Value
☒ app.dr-de...	A	Failover	Primary	No	13.232.214.252

STEP 16: Secondary

Route 53

- Dashboard
- Hosted zones**
- Health checks
- Profiles

▼ **Global Resolver**

- Global resolvers [New](#)

▼ **VPC Resolver**

- VPCs
- Inbound endpoints
- Outbound endpoints
- Rules
- Query logging

Records (1/4) Info

Automatic mode is the current search behavior optimized for best filter results. [To change modes go to settings.](#)

Filter records by property or value | Type | Routing p... | Alias

Record ...	Type	Routin...	Differ...	Alias
☐ dr-demo.c...	SOA	Simple	-	No
☐ app.dr-de...	A	Failover	Primary	No
☒ app.dr-de...	A	Failover	Secondary	No

Record details

[Edit record](#)

Record name
app.dr-demo.com

Record type
A

Value
13.219.237.170

Alias
No

TTL (seconds)
60

STEP 17: Test Disaster Recovery

- Simulate failure. ➤ Go to EC2 → Mumbai instance. ➤ Click:
- Stop Instance

EC2

- Dashboard
- AWS Global View
- Events

▼ **Instances**

- Instances**
- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations
- Capacity Manager [New](#)

Instances (1/1) Info

Successfully initiated stopping of i-0be34176cd334a47b

Last updated less than a minute ago | [Connect](#) | [Instance state](#) | [Actions](#) | [Launch instances](#)

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availabil
multi-region-server	i-0be34176cd334a47b	Stopped	t3.micro	-	View alarms +	ap-south

i-0be34176cd334a47b (multi-region-server)

[Details](#) | [Status and alarms](#) | [Monitoring](#) | [Security](#) | [Networking](#) | [Storage](#) | [Tags](#)

▼ **Instance summary** Info

Instance ID i-0be34176cd334a47b	Public IPv4 address -	Private IPv4 addresses 172.31.1.70
IPv6 address	Instance state Stopped	Public DNS

Now the primary server is down.



This site can't be reached

Check if there is a typo in app.dr-demo.com.

If spelling is correct, [try running Windows Network Diagnostics](#).

DNS_PROBE_FINISHED_NXDOMAIN

Reload

STEP 18: Now open backup-server

The screenshot shows the AWS Management Console interface for EC2 Instances. The left sidebar contains navigation links: EC2, Dashboard, AWS Global View, Events, Instances (selected), Instance Types, and Launch Templates. The main content area is titled 'Instances (1/1) Info' and includes a search bar, a table of instances, and action buttons like 'Connect', 'Instance state', 'Actions', and 'Launch instances'.

☒	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
☒	backup-virgini...	i-Off5a6cd9a73dba8f	Running	t3.micro	3/3 checks passed	View alarms +	us-east-1a

STEP 19: This confirms failover works.

Backup Server - Virginia

Conclusion:

The Multi-Region Disaster Recovery project successfully demonstrates a highly available and fault-tolerant cloud architecture using AWS services. By deploying resources across primary and secondary regions, the system ensures continuous application availability even during regional failures. Data replication using RDS and automatic failover through Route 53 guarantees minimal downtime and data consistency. The project highlights the importance of cloud resilience, scalability, and disaster preparedness in modern applications. Overall, it provides a reliable solution for maintaining business continuity and improving system robustness in real-world cloud environments.